AirPods Pro 3 × Galaxy S25 Ultra

The two remaining UWB acquisition blockers
Evidence, controller interfaces and bounded experiments

Research date: 9 October 2026 · **Target:** A3122 / 9A348 ↔ SR200D_A2_PROD / 1.64.4 / vendor UCI 3.0.2

SM-S938U · Android 17 / One UI 9 · KernelSU · LibrePods · Internal Bluetooth and UWB

Finding

Your controlled acquisition is a meaningful starting point, but the public material does not yet establish a working SR200D path for either an arbitrary Apple STS seed or the complete timed SP3-to-SP0 exchange. The strongest next step is to qualify the controller's timestamp, CRC and scheduled-transmit behavior while independently resolving ROSE's extra crypto operand.

There is one consequential correction: the IEEE D8 Annex H preset referenced by the related SR150T document uses a different key and counter from the MathWorks vector whose key matches Apple. E503=1 is therefore not a demonstrated shortcut to Apple acquisition STS. [S2–S3, S10]

Question	Research answer
ROSE FIFO 0x16 / 0x26?	No independent decoder of this exact pointer-based interface was located. Public Apple application-processor AES decoders use a different command dialect. [S6]
Logical STS counter?	64 + 32 + 32 is independently corroborated as a standard-shaped layout. IEEE fixes its own encrypt/increment and bit ordering; none of that assigns ROSE's extra scalar. [S2, S4–S5]
Arbitrary SR200D STS input?	Unproved. RFTEST presets, static fields and provisioned session keys are different mechanisms; acceptance/readback cannot establish the effective PHY key and counter. [S7, S10]
3 ms custom controller reply?	A plausible hardware capability, but no public complete implementation was found. RMARKER_TX, MAC_CFG and historical single-TX/test-ranging dispatches are concrete inspection leads. [S7–S11]
Existing non-Apple finder?	No inspected project demonstrates an Android/internal-radio replacement for the Apple finder with these exact STS and response details. Ghost Peak is a different experiment. [S13–S15]
Smallest decisive tests?	Fresh RX timestamps; one-packet CRC/loopback matrix; delayed TX in a validated clock domain; then a case-accepted reply. STS needs a real output oracle or a qualified arbitrary-seed path.

Scope and evidence boundary

This report builds on the supplied reverse engineering and RF observations; it did not operate the phone, case or Apple firmware. Only the IEEE AES known-answer calculation was independently executed offline. All proposed radio experiments use the phone's internal hardware, preserve ownership/enrollment and retain the existing bounded stop/deinitialize/Bluetooth-restoration path.

Source IDs resolve to the linked evidence ledger on pages 9–11. Page numbers cited for PDFs are printed page numbers unless a PDF index is explicitly supplied.

1. What is established—and what is not

State	Evidence and consequence
Supplied fact	The 128-event, 30 ms start predicts 3.840 s; acquisition/synchronization/SFD activity appeared at about 3.842 s, versus none in the unstarted baseline. This strongly associates the burst with the scheduled case transmission. STS-found remained zero. [S1]
Supplied fact	ROSE’s acquisition call submits the fixed key, address-derived 16-byte IV and independent scalar shown on the next page. Genuine 8B34 and 9A348 produce the same six-word submission under original-instruction execution. [S1]
Independently supported	IEEE STS uses AES-generated bits and a defined counter progression. A 64-bit upper field plus two 32-bit fields appears in Samsung’s patent and Qorvo’s FiRa implementation. [S2, S4–S5]
Plausible hypothesis	ROSE combines M0, scalar 0x2056C539 and a block counter into that shape. Placement, scalar byte order, the first block and hardware output packing remain open.
Unresolved on target	Whether SR200D uses the supplied static/session parameters in RFTEST; whether RX timestamps remain meaningful on STS failure; whether custom TX gains a CRC; and whether SP3 RX can trigger a correctly scheduled SP0 reply.

Two test-vector families must remain separate

Source	AES key K and first input V
MathWorks HRP configuration [S3]	K 4a5572bc90798c8e518d2449092f1b55 V 68debd3a599939dd57fdbb0e2a10fac0 Its key matches the supplied Apple key; its input does not.
IEEE D8 Annex H.2 [S2]	K 14148674d1d336aaf86050a814eb220f V 362eeb34c44fa8fbd37ec3ca1f9a3de4 This is the family explicitly referenced by SR150T E503.

The IEEE published first two output blocks are:

```
V0 = 362eeb34c44fa8fbd37ec3ca1f9a3de4
B0 = 7aa6f63ef917ae47115eb6fe3b5a5791
V1 = 362eeb34c44fa8fbd37ec3ca1f9a3de5
B1 = 41da0c7503566357ebf38b2c12bb3e92
```

Offline AES-128-ECB reproduced both blocks exactly: encrypt the supplied input first, then increment its final 32-bit word. IEEE defines upper96 plus counter32, MSB-first use of the generated bits and the pulse-polarity mapping. The further 64/32 partition comes from higher-layer implementations. These settle the standard algorithm, not ROSE’s input composition or memory packing. [S2, S4–S5]

Implication: if SR200D follows the SR150T preset literally, E503=1 selects a key and input that both differ from Apple. Under an M0-upper64 hypothesis, choosing RoseMAC cannot make the case use either published upper64: M0 begins 00, while those vectors begin 36 and 68. This is not a viable positive-control trick.

2. ROSE: the unresolved operand

Recovered submission, as supplied

```
FIFO address: 0x44005020
0x16000000   keyPointer
0x26020000   ivPointer
0x2056C539   outputPointer

K  = 4a5572bc90798c8e518d2449092f1b55
IV = M0[8] || 0000000000000000
M0 = 00 || RoseMAC[1:8]
```

Function 0x42BD4C submits these words and waits for the crypto engine. The length encoding is consistent with 512 output bytes. Generic function 0x41FFA0 adjusts the copied IV's final big-endian word by offsetBytes/16, chunks at 1024 bytes and advances that word by chunkBytes/16. It passes the extra scalar separately and unchanged. [S1]

What the independent code actually contributes

Qorvo's pinned FiRa implementation builds a PHY input by copying upper64, then writing a big-endian crypto STS index and a big-endian per-block counter. It also returns a derived authentication key, not automatically the original session key. Samsung's patent depicts the same 64/32/32 partition and maps VENDOR_ID to the low 16 bits of upper64, STATIC_STS_IV to its other 48 bits. Patent bit positions do not establish UCI byte order. [S4–S5]

Asahi m1n1 and PongoOS document application-processor AES engines with inline key/IV material and a separate crypt operation. Their superficial 1/2 opcode similarity does not decode ROSE's pointer-based 0x16/0x26 sequence. Substituting those decoders would introduce an unsupported interpretation. [S6]

Model	Candidate AES input, for output block i	Status
H1	`M0 || BE32(S) || BE32(i)`	Best standard-shaped starting candidate
H2	`M0 || LE32(S) || BE32(i)`	Scalar word may be loaded in a different order
H3	`M0 || zero32 || BE32(S+i)`	Scalar could seed the low counter
Variant	First i = 1 rather than 0	IEEE says 0 for its supplied V; ROSE behavior still needs evidence

Here S = 0x2056C539. H3 is not ruled out by the generic offset logic: adding an offset to the IV low word can commute with adding a scalar base. There is no evidence yet for arbitrary XOR, key whitening or a specific reseed operation; do not expand the search to such models without a firmware reason.

Smallest useful oracle

Obtain two consecutive **real crypto-engine output blocks**, plus the output for offsetBytes=16, with exact input buffers and scalar. After testing plausible memory packing, AES-decrypt the blocks under the known key; the recovered plaintexts can distinguish the three layouts and increment order. Repeat with a second controlled M0; observe a second scalar only if an authorized reachable caller supplies one.

An emulator that fills the output buffer using the assumed AES model is circular evidence. The supplied original-instruction runs establish CPU submissions, not the accelerator's result. First trace outputPointer consumers and existing owner-accessible diagnostics; no case memory-read or crypto-output command was established by this research. If none is reachable, this oracle remains blocked under the present interfaces.

3. What the Samsung controller must expose

Required capability is an effective PHY key plus input

A successful SET_CONFIG reply is insufficient. The useful capability is control over the exact AES key and 128-bit input, or control over an equivalent generated STS sequence, at the radio. Nothing inspected proves that arbitrary STS waveform RAM or raw STS bits are host-writeable on SR200D.

Interface	Useful evidence	Remaining issue
E5/03 RFTEST preset	SR150T describes selection of both key and DRBG input, proprietary versus D8 Annex H. [S10]	What SR200D's two presets contain; whether any fields override them.
02 / 0A / 27 / 28	STS_CONFIG, STS_INDEX, VENDOR_ID and STATIC_STS_IV give an apparent route to the logical fields. [S1, S4, S7]	Whether exact RFTEST consumes them and which effective key/low counter it uses.
45 SESSION_KEY	Public SR250 header limits this parameter to provisioned modes 3/4 and says other modes ignore it. [S7]	Cannot infer a raw AES-key override for static RFTEST.
Provisioned STS	Qorvo code derives authentication material from session inputs. [S5]	A supplied session key can differ from the final STS key; SR200D derivation is unverified.
Vendor / diagnostic path	Concrete dispatch and timestamp leads exist in related code. [S7–S11]	No public command found that simultaneously proves arbitrary Apple STS and timed custom reply.

Keep four generations distinct

Target / provenance	What can be carried forward
SR200D: actual phone	Your successful bounded acquisition and RFTEST observations are the target evidence. Firmware 1.64.4 and vendor UCI 3.0.2 must anchor all new traces. [S1]
SR150T: rev. 2.0.17 manual	Useful related definitions for E503, diagnostic timestamps and MAC_CFG; it is explicitly an SR150T document. It does not certify SR200D behavior. [S10]
SR250: public host SDK	Exact serializers, enums and structures provide candidate bytes and inspection locations. The repository targets SR250 middleware, not the S25 controller binary. [S7–S9]
SR100T: original 2023 research	Galaxy S21 Ultra / Android 11 / fw21.10.0 / UCI1.10; historical TEST_TX and TEST_SS_TWR declarations are leads to inspect, not portable working commands. [S11]

One implementation difference is concrete: the SR100T research lists TEST_TX at GID 0D/OID 04; the inspected SR250 test header reserves 04. Historical RAW_PHR at test parameter 05 is two bytes; SR250 defines a one-byte PHR_RANGING_BIT there. Reusing old field lengths or notifications can silently misparse results. [S8, S11]

4. Concrete command and CRC leads

The following bytes have been checked against the public SR250 serializers and routing. They are **candidates for verification on SR200D**, not claims of support. They assume the already initialized session/handle 0, normal four-byte UCI framing, no fragmentation and no extra transport wrapper. [S7–S9]

```
GET MAC_CFG
2F 03 00 06  00 00 00 00  01  70

SET MAC_CFG = 02
2F 00 00 08  00 00 00 00  01  70 01 02

SET E503 = 00 / 01
2D 00 00 09  00 00 00 00  01  E5 03 01 00
2D 00 00 09  00 00 00 00  01  E5 03 01 01

SET RMARKER_TX
2D 00 00 0B  00 00 00 00  01  06 04 <LE32 target>

TEST_PERIODIC_TX with 23-byte payload
2D 02 00 17  <23 PSDU bytes>

TEST_LOOPBACK with 23-byte payload
2D 06 00 17  <23 PSDU bytes>

TEST_RX:   2D 05 00 00
TEST_STOP: 2D 07 00 00
```

No additional `17 00` PSDU-length prefix belongs inside those 23 payload bytes. The builder does not add a UCI checksum. The over-air CRC is a different question. Configure one packet through the existing validated NUM_PACKETS path; a periodic-TX command need not mean an unbounded test.

MAC_CFG is the most concrete new CRC lead

The SR250 header defines vendor parameter 0x70: bit 0 = MAC header present, bit 1 = MAC footer present; defaults are 3 for FiRa and 0 for test sessions. The term “present” does **not** say whether firmware generates, expects, checks or strips the footer. Comparing values 0 and 2 is therefore an experiment, not an established append-CRC switch. [S7]

The exact custom payload

```
41 A8 || seq[1] || M0[8] || M1[8] || A5 A5 || 00 || 31
M0 = 00 || RoseMAC[1:8]
M1 = 01 || RoseMAC[1:8]
Total supplied PSDU = 23 bytes; expected with CRC16 = 25
```

Treat these as opaque bytes. Do not normalize them into a standard MAC header or allow a generic FiRa encoder to replace addresses, PAN ordering or the message kind. The source establishes the desired bytes and case validator length, but Samsung’s CRC addition remains unproved. [S1]

Use a one-packet SP0 loopback matrix: 23 versus 25 bytes with an independently computed FCS, crossed with MAC_CFG 0 versus 2. Inspect decoded PHR length, returned PSDU length, CRC status and any returned footer. A 23-byte RX payload alone is inconclusive if hardware strips FCS; a successful loopback alone is also inconclusive if it bypasses CRC.

5. The 3 ms problem is a controller problem

Four capabilities are needed together

The path must receive the case's SP3 acquisition, identify the same timing reference used by the Apple reply schedule, select the SP0 transmit format and send a preloaded custom PSDU at the target time. Having each primitive in a host API does not establish that firmware can compose them within one receive window.

The reply itself has no STS. Consequently it may be possible to demonstrate case acceptance before solving STS, if SR200D supplies a prompt, fresh acquisition timestamp despite STS mismatch and can schedule the SP0 packet. Counter-only PER notifications or a TEST_RX notification emitted after a long timeout are insufficient.

Clock / field	Duration corresponding to 3 ms	What is not established
Apple recovered 48-bit schedule	115200 ticks, implying 38.4 MHz	The relation between this timestamp and Samsung's receive marker. [S1]
Related NXP RFTEST 124.8 MHz snapshot	374400 ticks = 0x0005B680	Exact SR200D units, RX/TX epoch, marker and valid-on-failure behavior. [S8, S10]
Android UWBS microsecond API	3000 microseconds	No proven mapping to RFTEST snapshot ticks or its wrap/epoch. [S20]

The numerical conversions in this table are calculations, not measurements on the S25. A 32-bit counter at 124.8 MHz wraps in about 34.4148 seconds. Only after validating that domain could one try `target = (rx + 374400) mod 2^32`.

RX_DONE, RMARKER and host receipt are different events

The related RX structure includes integer/fractional timestamp fields; SR150T diagnostics also describe a received RMARKER and STS-failure status. Those field names do not guarantee fresh values on an STS-failed SP3 frame. Validate freshness with shifted case starts and the known quiet baseline. [S8, S10]

Determine whether a TX timestamp reports the actual RF marker, end of transmission, a command-time snapshot or an echoed requested target. Varying the payload length helps distinguish a packet-end marker. The SP3 duration itself can create a material offset if one endpoint schedules from RMARKER and the other from RX completion.

Public interfaces do not yet show an Apple exchange sequencer

The public RFTEST entry point exposes periodic TX, PER RX, RX, loopback and security RX. Its structures contain scheduling-looking fields, but no documented RX-triggered arbitrary-reply template. NXP's public "P2P" example describes a standard ranging session and does not establish this custom acquisition protocol. [S8, S12]

Historical TEST_TX 0D/04 and TEST_SS_TWR 0D/08 are valuable dispatch-search terms. The thesis did not validate its RFTEST data commands on air; a standard single-sided ranging test also would not automatically accept Apple's bytes and SP3/SP0 transition. [S11]

Implementation consequence

Preload and arm as much as firmware allows. A host implementation that waits 3 ms after a callback has the wrong reference time. Even immediate host handling plus a hardware target is viable only if measured worst-case command/PHY-transition latency leaves enough lead time. Root access alone does not confer a controller scheduler or firmware modification capability.

6. Ranked bounded experiments: first gates

Use the established case session and Samsung cleanup path throughout. Keep packet counts finite, log raw commands/responses/notifications, use monotonic host timestamps and explicitly stop/deinitialize after each trial. These are proposed tests; no phone experiments were run for this report.

1 · Read-only dispatch audit, then a fresh RX timestamp

Audit: inspect the exact S25 command path and confirm candidate tags/opcodes before invoking unfamiliar commands. Capture firmware and host-component identities. Prefer one known TEST_RX session with the current PHY; use one unstarted baseline and three bounded case starts whose event counters shift the burst by known amounts.

Success: a per-packet timestamp follows the imposed shifts, is absent or marked invalid in the baseline, and arrives promptly enough to be useful for a 3 ms reply. Preserve the raw timestamp, status, integer/fraction fields and host receipt time; derive units only from observed differences and documented field meanings.

Failure/inconclusive: only aggregate counts; stale/zero timestamps; a notification only at timeout; or timestamps without a validity indication on STS failure. That blocks this host-mediated acquisition route, not all possible firmware-assisted routes.

2 · Resolve one-packet SP0 framing and FCS

After confirming the candidate vendor path, run the four loopback cases from page 5: PSDU length 23 or 25, MAC_CFG 0 or 2. Keep addressing bytes, PHY, sequence, transmit count and randomization fixed. The 25-byte alternative must contain a separately verified IEEE 802.15.4 FCS over the exact 23 bytes.

Success: a unique configuration produces the expected decoded 25-byte on-air length and explicit valid-FCS evidence, or returns the expected two CRC bytes. Repeat with one changed payload byte and its corresponding CRC to rule out a stale buffer.

Failure/inconclusive: accepted configuration without packet evidence, a stripped-FCS payload without CRC status, an internal path that skips framing checks, or a test rejected for an unrelated precondition. Do not alter permanent calibration solely to force a loopback to start. Successful internal loopback still requires confirmation against the case.

3 · Prove delayed transmit in one clock domain

First establish a readable scheduler counter or a validated mapping from the receive counter to the scheduling field. Then configure one SP0 packet and test safely future targets, for example 5 ms and 10 ms ahead when lead time permits. Change host submission lead time while holding each target fixed.

Success: an actual hardware TX timestamp tracks the requested target rather than host submission/completion time, including a documented wrap calculation. A bounded late-target trial should exhibit a consistent missed-deadline response rather than an unexplained immediate transmit.

Failure/inconclusive: RMARKER_TX merely echoes a snapshot, the field is ignored, the available epoch is unrelated, or the returned value is just the requested timestamp. Distinguish these before trying RX+3 ms.

What these gates buy

Experiments 1–3 isolate reception, frame generation and scheduling. They can establish a feasible custom-reply transport without assuming Apple STS is solved. Conversely, 5–10 ms standalone delayed TX does not prove that firmware can switch SP3-to-SP0 and meet the shorter receive-relative deadline.

7. Ranked bounded experiments: decisive gates

4 · One case-accepted reply at RX + 3 ms

Preload the exact 23-byte SP0 reply under the best framing result. Arm the timestamped acquisition path and schedule against the validated receive marker/domain. Keep the trial to one acquisition/reply opportunity and observe the case's recovered state machine.

Success: a unique, repeatable transition on the case's validated reply path or a subsequent protocol action that depends on that reply. Use the exact existing firmware trace to identify the strongest observable state. Before using acceptance as proof of FCS correctness, trace that successful validation is gated by a hardware CRC-good indication; the supplied 25-byte length check alone does not show this.

Controls: wrong M0 or message kind, and a target outside the known receive window, with deinitialization of the precision session between runs. A success should disappear under each relevant negative control. These controls preserve ownership and enrollment.

Failure/inconclusive: transmit-complete notification without case progression. Separate payload/FCS rejection, timestamp-reference offset, deadline miss and PHY switch failure rather than attributing all failures to STS.

5 · Resolve ROSE's scalar with the smallest real output oracle

In parallel with controller work, trace outputPointer consumers and already reachable diagnostics. If a legitimate read path exists, capture 32 bytes at offset zero and 16 bytes at offset 16, with input buffers/scalar. Compare candidate decryption layouts H1–H3, then repeat for a different RoseMAC.

Success: two consecutive outputs identify a coherent AES input/increment rule; offset 16 yields the same next block; the second M0 behaves as predicted. Confirm output packing before extending to the entire 512-byte STS buffer.

Failure/inconclusive: a software-filled output buffer, only repeated CPU FIFO submissions, or no reachable accelerator output. Do not call a counter construction recovered until the observations constrain it.

6 · Qualify effective SR200D STS inputs, then test the case

Follow the exact test-session selection to the effective key/counter loader or a diagnostic readout if visible. Readback or shared-preset loopback cannot qualify arbitrary seed control. If no such view exists, a finite case-based candidate probe is still useful, but only positive, differential STS evidence can qualify the path.

Test the small H1/H2/H3 set and justified first-block variants; use two RoseMAC values, disable or account for auto-increment and add a one-bit incorrect-input control. Keep packet count and PHY unchanged.

Success: repeatable STS success/quality and valid timestamps only for the predicted inputs, disappearing for the one-bit negative control. That distinguishes STS matching from the already demonstrated preamble/SFD reception.

Failure/inconclusive: all-negative results. They cannot distinguish an incorrect Apple model from ignored Samsung parameters, a preset key, a hidden KDF, output packing or a timing/PHY issue. A verified Apple buffer does not by itself prove Samsung can synthesize it.

Stop condition and next milestone

A case-accepted acquisition reply and validated acquisition STS are separate milestones. Full Precision Finding additionally needs the subsequent ranging exchange, valid secure timestamps, distance calibration and directional measurements. Advance only on those measurements; none of the acquisition counters or the proposed self-tests establishes distance or direction.

8. Evidence ledger · STS and crypto

ID / exact source	Location	Supported conclusion and limit
S1 · Supplied reverse-engineering and RF observations	ROSE 0x42BD4C, 0x41FFA0; FIFO 0x44005020; 8B34 and 9A348; scheduled case start	User-provided evidence: exact FIFO words/key/IV, offset handling, reply bytes and 3 ms schedule; controlled burst acquisition without STS. This report did not independently execute those firmware paths or RF trials.
S2 · IEEE P802.15.4z/D8 (original standard draft, public mirror) IEEE 802.15.4z-2020 (public mirror)	D8 Annex H.2 p184, H.3 p185; final Annex H.2 p171, H.3 p172; final §15.2.9.2 / Fig. 15-7b p146, Table 11-2 p138	Exact Annex H key/input/output; encrypt then increment; standard bit/pulse mapping. Two AES outputs independently reproduced. Mirrored original text was inspected, not the hosting site's generated summary. Does not document ROSE.
S3 · MathWorks lrwpanHRPConfig	STSPacketConfiguration property / STS test key and counter	Documents the 4a55... key and 68de...2a10fac0 input. The key matches the supplied Apple key, but this is not the same pair as D8 Annex H. No Apple-specific mapping.
S4 · Samsung US11936771B2	Figs. 5, 6 and 8 and associated detailed description; Fig. 5 PDF p7, Fig. 8 PDF p10	64-bit upper field / 32-bit crypto STS index / 32-bit counter; vendor and static-IV bit allocation. Describes an invention, not a recovered SR200D implementation or UCI byte order.
S5 · Qorvo DW3000 FiRa fira_crypto.c commit 11a0813757dbcc9dd4818a807a8d000eb4766916	fira_crypto_context_init, lines 324–384; fira_crypto_get_sts_params, lines 538–574; key rotation helpers	Independent code builds upper64 || BE32(index) || BE32(counter). Static seed and provisioned session inputs feed KDFs; output PHY key is derived. Evidence about Qorvo's stack, not ROSE or SR200D.
S6a · Asahi m1n1 hw/aes.py	AES_OPCODE, AESSetKeyCommand, AESSetIVCommand, AESCryptCommand	Application-processor command dialect: KEY=1, IV=2, CRYPT=5, with field definitions. It does not independently decode ROSE's six-word pointer submission.
S6b · PongoOS aes_a9.c	AES operation submission, lines 153–197	Inline key words and IV words, followed by a separate data command and addresses. Reinforces that similar Apple opcode prefixes do not establish the ROSE format.

Negative findings are bounded: no exact ROSE FIFO documentation was located among the inspected sources. This is not proof that no vendor/internal documentation exists.

9. Evidence ledger · NXP interfaces

ID / exact source	Location	Supported conclusion and limit
S7 · NXP UwbApi_Types_Proprietary.h UwbApi.c	Header lines 587–598: SESSION_KEY; 927–935: MAC_CFG. UwbApi_SetStaticSts 1130–1175; SetVendorAppConfigs 812–876; GetVendorAppConfigs 1040–1074	SR250 host definitions: key mode restrictions; 0x70 header/footer presence and defaults; vendor/static setter construction. These describe host paths, not measured SR200D radio behavior.
S8 · NXP UwbApi_RfTest.c UwbApi_Types_RfTest.h uci_test_defs.h	UwbApi_StartRfTest 442–507; phRfTestParams_t 70–100; RX/loopback structures 170–249; command definitions 26–55	SR250 enums for 0D/02,03,05,06,07,09; 04 reserved. RMARKER TX/RX fields described as 124.8 MHz snapshots; timestamp structures. No exposed arbitrary RX-triggered reply sequencer.
S9 · NXP UwbApi_Utility.c UwbApi_Internal.c uci_hmsgs.c	serializeAppConfigPayload 90–100; serializeTestDataPayload 132–137; app TLVs 1802–1863; test TLVs 1870–1905; getGidOid 204–208; uci_snd_cmd 284–380	Exact session/count, integer little-endian and byte-array serialization; PSDU copied unchanged; four-byte UCI header. Derivation of candidate wire examples, not target support.
S9b · NXP proprietary serializer uci_ext_defs.h uci_defs.h	getVendorAppConfigTLVBuffer 163–225; getExtTestConfigTLVBuffer 295–321; vendor OIDs 304/307; GID 0F at uci_defs.h94; uwb_int.h75,346,348	Vendor SET/GET routing 0F/00 and 0F/03; MAC_CFG 70/01/value; E5/03/01/value. All candidate payloads fit one unfragmented control packet.
S10 · NXP_SR150T_UCI_Specification rev. 2.0.17, 7 Feb 2025 (mirror)	§11.11 Table 142 p122; §8.2 Table 71 pp56–57; §9.3 Tables 81–82 pp62–63; Table 123 p109	E503 selects both key and DRBG input and explicitly cites D8 Annex H. RX integer/fraction, diagnostic RMARKER/status and related MAC_CFG definitions. Document says SR150T and is marked CONFIDENTIAL; not an SR200D specification.
S11 · Martin Reza Heyden, 2023 thesis: Security Analysis of Samsung's UWB Ecosystem	§7.3.2 pp61–63; Table 51 p127 / PDF145; Table 52 p128; Table 55 p130; Table 129 p158; Tables 216–222 p174	SR100T/S21 Ultra firmware context; TEST_TX 0D/04 and TEST_SS_TWR 0D/08 declarations; old RX/RAW_PHR formats. RFTEST data commands were not actually tested in the thesis.
S12 · NXP UWBIoT Middleware Guide	§2.4.3 p12; §6.1 p43; historical SR100T/SR040 example	P2P example is a standard ranging session (type 00). The name supplies no evidence of Apple-compatible SP3 reception followed by a custom SP0 response.

10. Evidence ledger · implementations and access

ID / exact source	Location	Supported conclusion and limit
S13 · Leu et al., Ghost Peak, USENIX Security 2022 Authors' project page	§3.3, PDF p9; §3.4 interoperability evaluation	Reactive attack/sniffer and legitimate Apple/NXP NI pairing are separate roles. AirTag ranging retains an iPhone as legitimate finder; the work does not recover this Apple acquisition seed or implement an Android case finder.
S14 · SEEMOO apple_u1	README; rose_sniff_kernel.js; accompanying U1 research material	Useful host/AOP/IOKit tracing and research tooling. Not an independent ROSE 0x16/0x26 decoder or an owner-accessible AirPods crypto-output interface.
S15 · SEEMOO uwb-sniffer	README reception/configuration caveats; timestamp/frame reception implementation	Packet observations and configurable sniffing are not matched secure STS. Forwarded malformed/incorrect-length frames reinforce the need for validity checks. Not a replacement precision finder.
S16 · Apple Nearby Interaction Accessory Protocol specification endpoint UWB Interoperability specification endpoint	Technology/documentation page and current specification links	Specification endpoints redirect to Apple Developer sign-in. Additional entitlement or NDA terms were not verified. The published program does not by itself document this legacy owned-case protocol.
S17 · NXP SR250 documentation catalog	UM12508, SR250 UCI Specification, rev. 1.0, 16 Mar 2026	Catalog labels the current detailed manual Account Required. This is an SR250 document, not the exact SR200D OEM command specification.
S18 · NXP official support: UWB MAC Custom Session	NXP TechSupport response, 19 Sep 2024	Support says detailed UWB information is under NDA. This supports requesting vendor access; exact SR200D terms/authorization remain unverified. User-posted parameter claims are not NXP confirmations.
S19 · FiRa specifications resource hub	Specification access and purchase options	Specifications are available to members or by purchase. Public host headers are not a substitute for the relevant UCI Test specification.
S20 · AOSP UwbManager.java Pinned UwbSessionManager.java	queryUwbsTimestampMicros; computeAbsoluteInitiationTime	Framework paths use a UWBS microsecond timestamp. This does not define the RFTEST snapshot domain, marker or mapping to the Samsung CORE08 handling observed by the user.

11. Access requests and LibrePods handoff

Document / interface	Access status	What to ask for
Apple Nearby Interaction specifications	Current endpoints require Developer sign-in; extra entitlements/NDA unverified. [S16]	Clarification whether any document covers legacy acquisition or ROSE accelerator descriptors, rather than only current NI interoperability.
NXP SR200D OEM UCI / RFTEST	No exact public manual located for fw1.64.4 / vendor UCI3.0.2. NXP support describes detailed UWB access under NDA. [S18]	The exact firmware's effective STS seed selection, RX timestamp validity on STS failure, 0x70 FCS semantics, RMARKER_TX epoch and RX-triggered SP3→SP0 scheduling.
NXP SR250 UM12508	Account Required in official catalog. [S17]	Useful related schema; confirm which features exist in SR200D before carrying any command across.
SR150T rev. 2.0.17	Public mirror of a document marked CONFIDENTIAL. [S10]	Use as qualified related evidence; it is not an official public release or an SR200D contract.
IEEE / FiRa	IEEE text inspected via mirrors; original Mentor contribution download failed with HTTP418. FiRa offers member/purchase access. [S2, S19]	Obtain authoritative test-vector files and matching UCI Test revision. A retrieval error does not establish login or NDA restrictions.
ROSE crypto FIFO manual	No independently decoded public interface located. [S6]	Exact accelerator command/operand description, or a real output trace for the recovered acquisition call.

Minimal LibrePods research implementation

Keep the proven root transport and owner BLE session flow. Add a bounded research runner that records each raw UCI command, response and notification alongside firmware identity, PHY settings, RoseMAC, start event and monotonic receipt time. Store timestamp bytes before assigning units or labels, and record whether each value is fresh and valid for the observed status.

Separate experimental results into four explicit states: **acquisition observed**, **STS validated**, **reply accepted by the case** and **ranging/direction validated**. Associate every promotion with its positive and negative controls. This prevents a configuration readback, a transmit completion or an SFD counter from being reported as Precision Finding.

Use one packet or one acquisition opportunity per initial test, with the existing watchdog, test stop, session deinitialization and Android Bluetooth restoration. Parameter sweeps should be finite and justified by the candidate models; do not start by sweeping unknown vendor opcodes or undocumented register writes.

Decision after the first three experiments

If fresh RX timestamps, correct custom framing and delayed SP0 transmission all work, the next concrete target is a case-accepted reply at the validated RX-relative time. If an exact missing primitive is identified, focus the firmware audit or vendor question on that primitive. If no effective arbitrary STS input is exposed, recovering the Apple counter alone will not make the internal radio reproduce it.